Credit Risk Analysis

Exploratory Data Analysis Report

Prepared by: Mehedi Hasan
Date: May 2026
Dataset: Credit Risk Dataset (Kaggle)
Total Records: 32,574 Loans

T

Table of Contents

1. Executive Summary

This report presents an exploratory data analysis of a credit risk dataset containing 32,574 loan records. The dataset includes borrower demographics, loan characteristic, and payment behavior intent and grade.

The analysis reveals that approximately 21.8% of loans resulted in default. Key drivers of default include loan grade, income level, previous default history, and interest rate. High risk grade loans (E, F, G) show a staggering 67.3% default rate while low income borrowers default at nearly twice the rate of high income borrowers.

These findings suggest that loan grade, borrower income, and previous default history should be the primary factors considered in credit risk assessment and loan approval decisions.

2. Data Overview

2.1 Dataset Source

The dataset was obtained from Kaggle (Credit Risk Dataset by Laotse). It contains 32,574 rows and 12 columns representing loan applicants and their repayment behavior.

2.2 Key Variables

- person_age: Applicant's age
- person_income: Annual income
- person_home_ownership: Home ownership status
- person_emp_length: Employment length in years
- loan_intent: Purpose of the loan
- loan_grade: Credit grade assigned to the loan
- loan_amnt: Loan amount requested
- loan_int_rate: Interest rate on the loan
- loan_status: Whether loan defaulted (1) or not (0)
- loan_percent_income: Loan amount as % of income
- cb_person_default_on_file: Previous default history
- cb_person_cred_hist_length: Credit history length

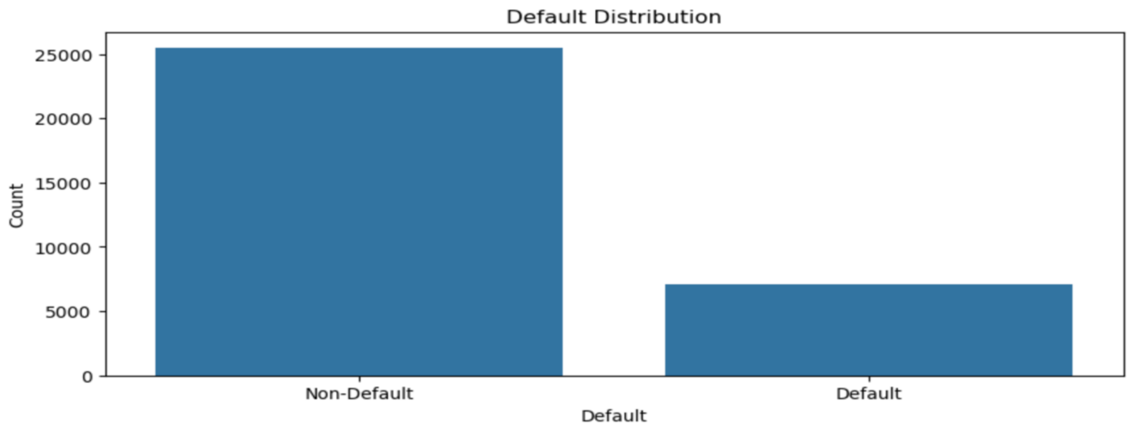
2.3 Data Cleaning Steps

- Removed age outliers (age > 100)
- Removed emp_length outliers (emp_length > 60)
- Capped person_income at 99th percentile ($225,000)
- Filled missing loan_int_rate with median grouped by loan_grade
- Filled missing emp_length with median
- Created label column for loan_status
- Created age_group, income_group, loan_group,
- interest_rate_group, loan_risk bins

3. Exploratory Data Analysis

3.1. Univariate Analysis

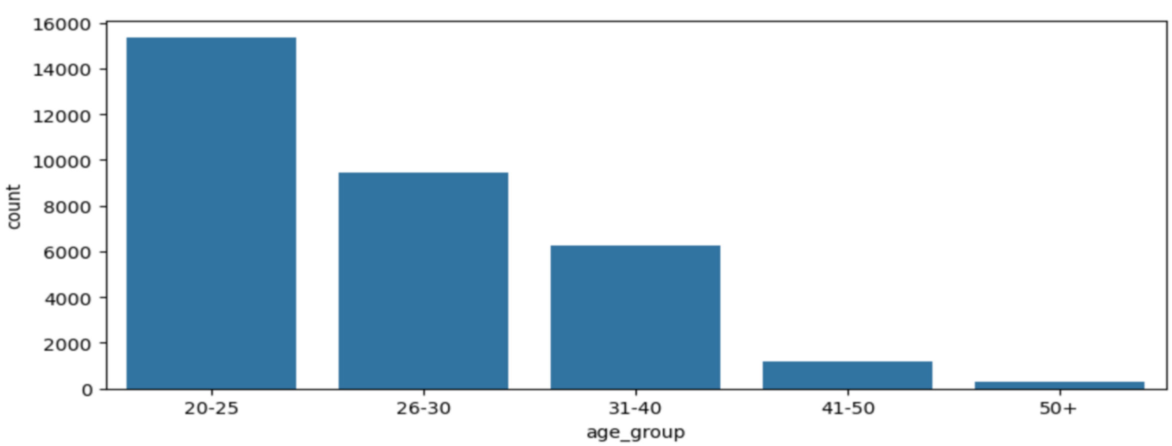
1. What is the overall default rate?




```
loan_status_label
Non-Default    78.178077
Default        21.821923
```

Approximatly 21.8% of loan applicants defaulted, while 78.2% successfully repaid. This indicates a class imbalance in the datasets, similar to churn analysis.

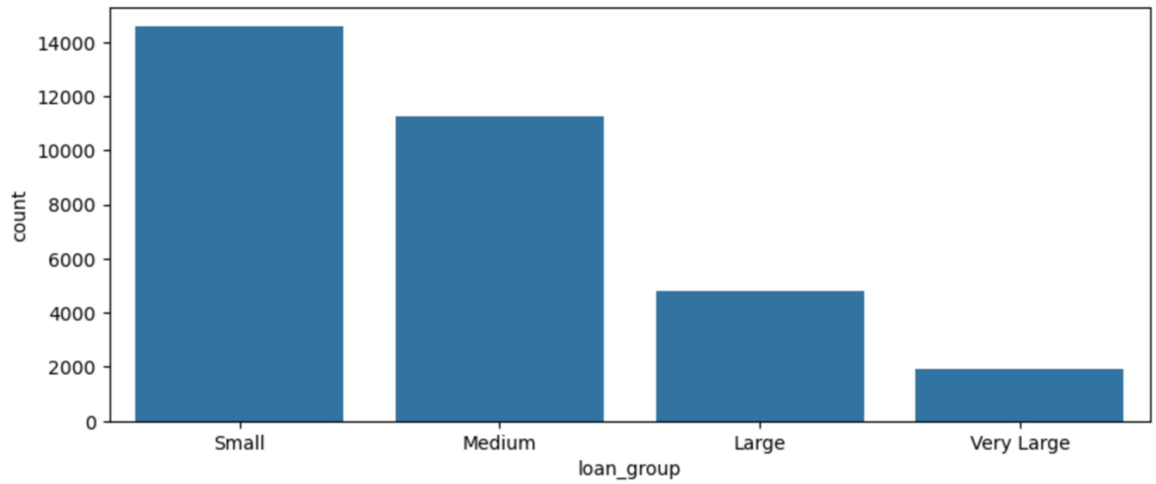
2. What is the age distribution of loan applicants?




```
age_group
20–25    47.099112
26–30    29.091803
31–40    19.235849
41–50     3.704045
50+       0.869191
```

Majority of loan applicants are young 47.1% are aged 20-25, followed by 26-30 (29.1%). This suggests that the dataset is dominated by young borrowers, possibly students or early career professionals.

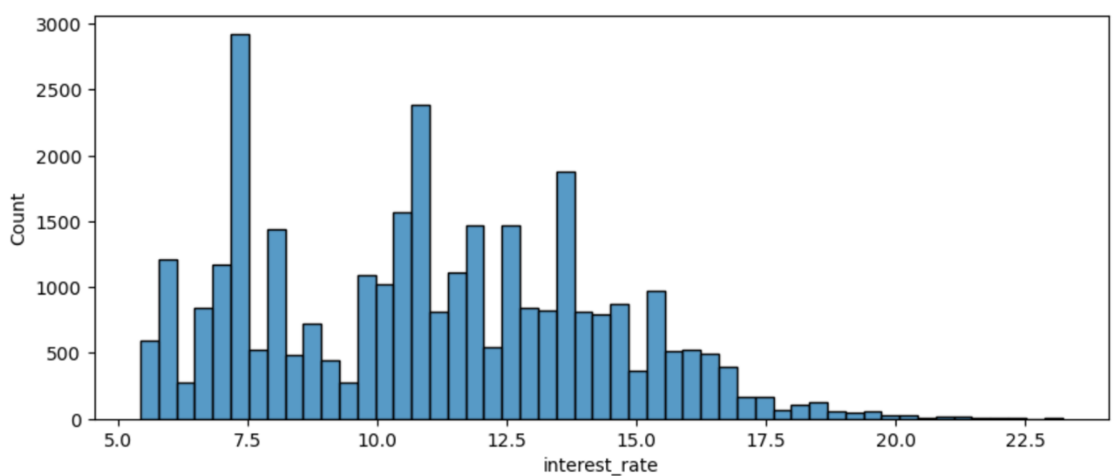
3. What is the distribution of loan amounts?




```
loan_group
Small          44.749532
Medium         34.617157
Large          14.699469
Very Large      5.933843
```

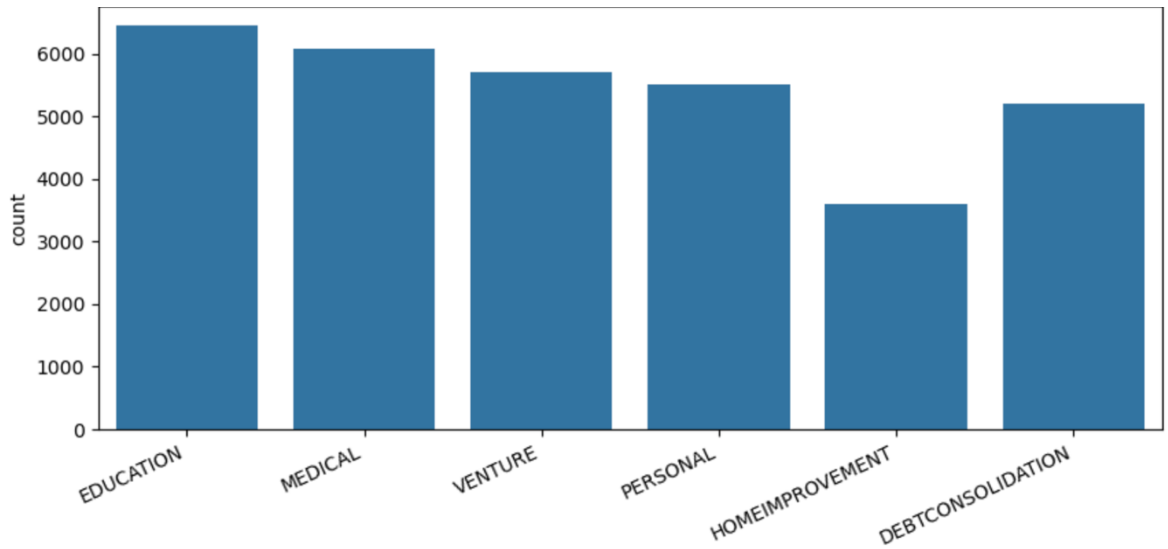
Majority of loans are Small (44.8%) and Medium (34.6%). Very Large are rare (5.9%), suggesting most applicants borrow relatively modest amount.

4. What is the distribution of interest rates?



Interest rate distribution is multimodal with peaks around 7.5%, 10.5%, and 13.5%. This suggests different loan grades have distinct interest rate ranges. Most loans fall between 7-15% interest rate.

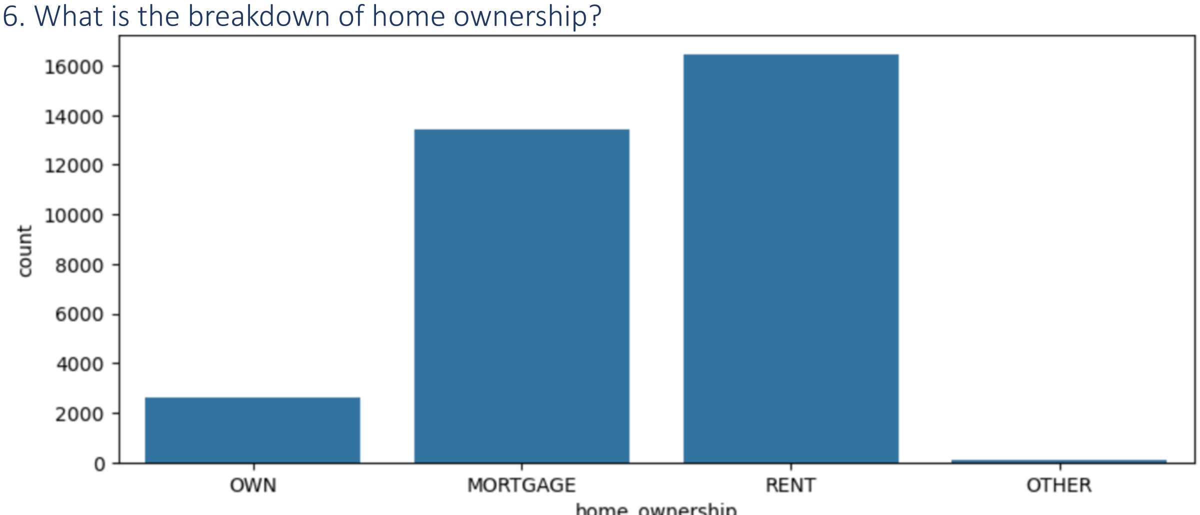
5. What is the breakdown of loan intent?




```
loan_intent
EDUCATION            19.797905
MEDICAL              18.643079
VENTURE              17.540465
PERSONAL             16.944624
DEBTCONSOLIDATION    16.001720
HOMEIMPROVEMENT      11.072207
```

Loan intent is fairly evenly distributed. Education (19.8%) and Medical (18.6%) on the top reasons for borrowing followed by Venture (17.6%) and Personal (16.9%). Home Improvement has the lowest share (11.1%).

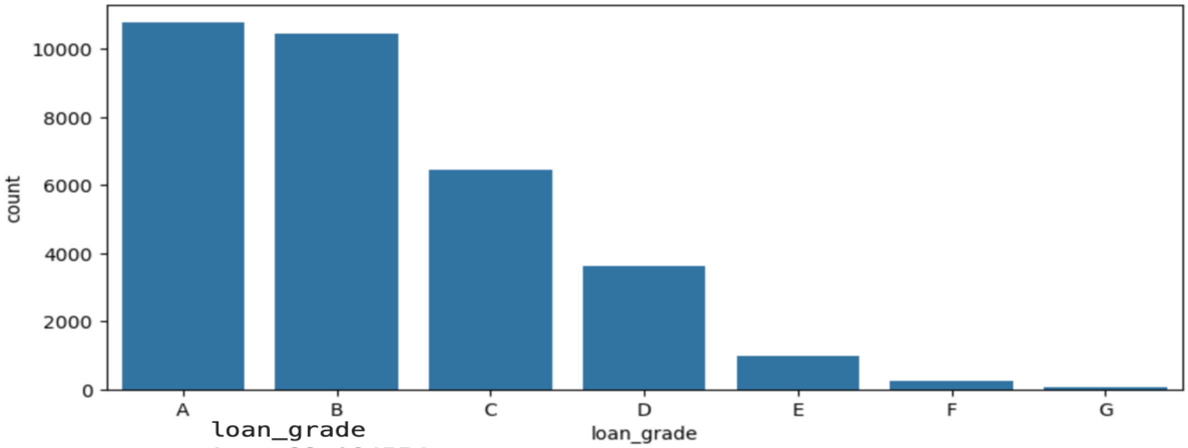
6. What is the breakdown of home ownership?




```
home_ownership
RENT        50.486809
MORTGAGE    41.248196
OWN          7.936362
OTHER        0.328634
```

Half of all applicants are renters (50.5%), followed by mortgage holders (41.3%). Only 7.9% own their home outright. This suggests most of applicants do not have strong asset backing for their loans.

7. What is the breakdown of loan grade?

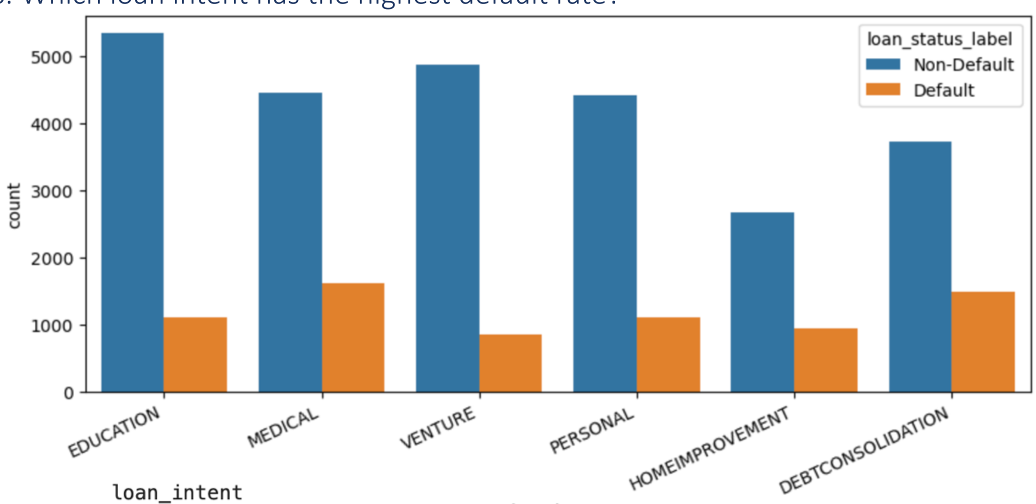



```
loan_grade
A      33.084554
B      32.074081
C      19.822476
D      11.121349
E       2.960779
F       0.740195
G       0.196566
```

Majority of loans are Grade A (33.1%) and Grade B (32.1%), indicating most applicants have good creditworthiness. High risk Grade E, F, G combined only 3.9% - very few applicants fall in the extreme risk category.

3.2. Bivariate

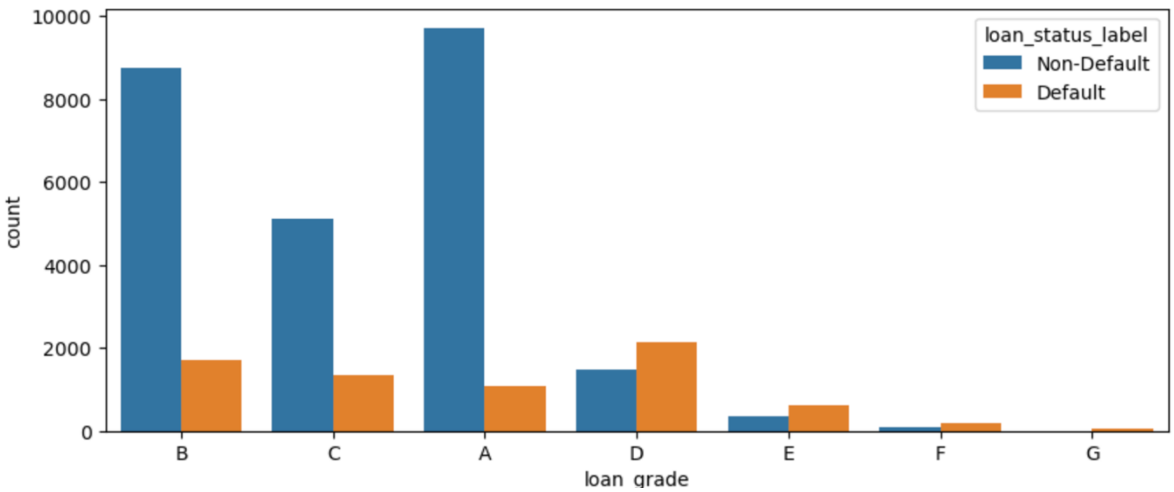
8. Which loan intent has the highest default rate?




```
loan_intent
DEBTCONSOLIDATION      28.598848
EDUCATION              17.235495
HOMEIMPROVEMENT        26.102635
MEDICAL                26.705107
PERSONAL               19.865869
VENTURE                14.813518
```

Debt Consolidation loans have the highest default rate (28.6%), followed by Medical (26.7%) and Home Improvement (26.1%). Venture loans have the lowest default rate (14.8%), suggesting business oriented borrows are more financially disciplined.

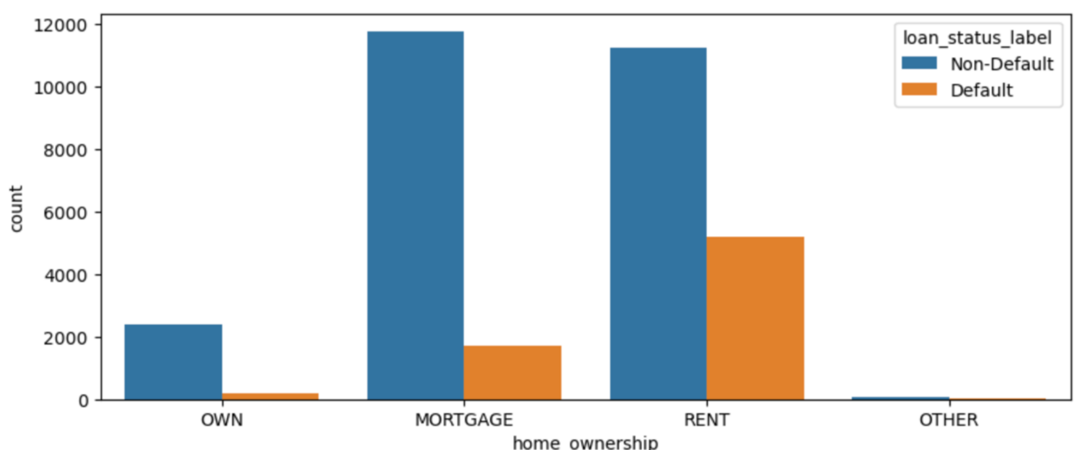
9. Which loan grade has the highest default rate?




```
loan_grade
A     9.961010
B    16.288423
C    20.746824
D    59.044463
E    64.419087
F    70.539419
G    98.437500
```

Default rate increase dramatically with loan grade. Grade A has only 9.9% default rate while Grade G has 98.4% - almost certain default. This confirms loan grade is the strongest predictor of default risk. Grade D, E, F, G should be considered extremely high risk.

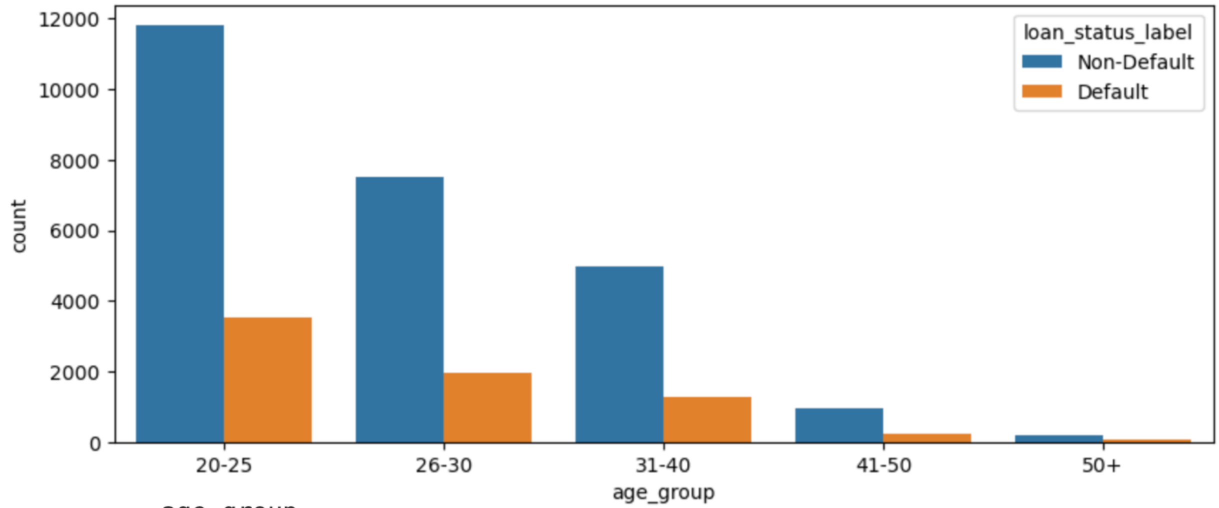
10. Does home ownership affect default rate?




```
home_ownership
MORTGAGE    12.576322
OTHER       30.841121
OWN          7.469040
RENT        31.573184
```

Renters has the highest default rate (31.6%), followed by Other (30.8%). Home Owners have the lowest default rate (7.5%). Suggesting that owning a home indicates better financial stability and lower credit risk.

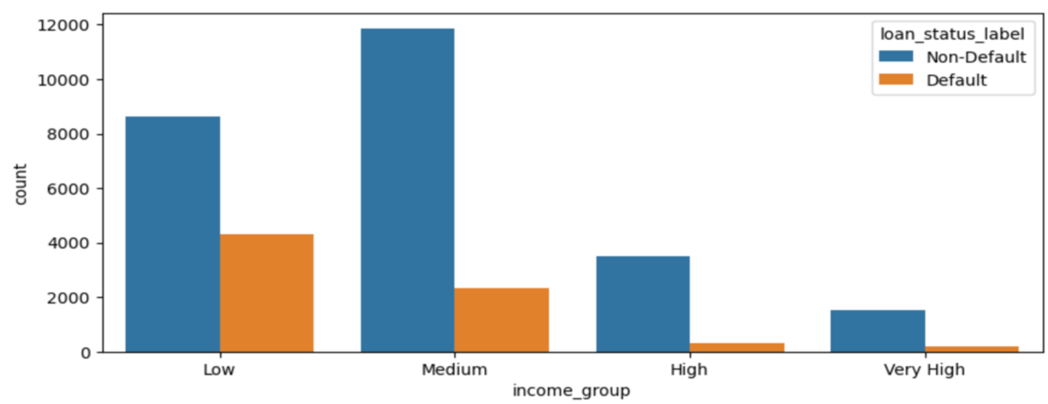
11. Which age group defaults the most?




```
age_group
20-25     23.032279
26-30     20.882601
31-40     20.373623
41-50     20.398010
50+       25.795053
```

Default rate is fairly consistent across age group (20-25%). Youngest (20-25) and oldest (50+) borrowers default slightly more. However, since 76% of applicants are under 30, younger borrowers represents majority of actual default.

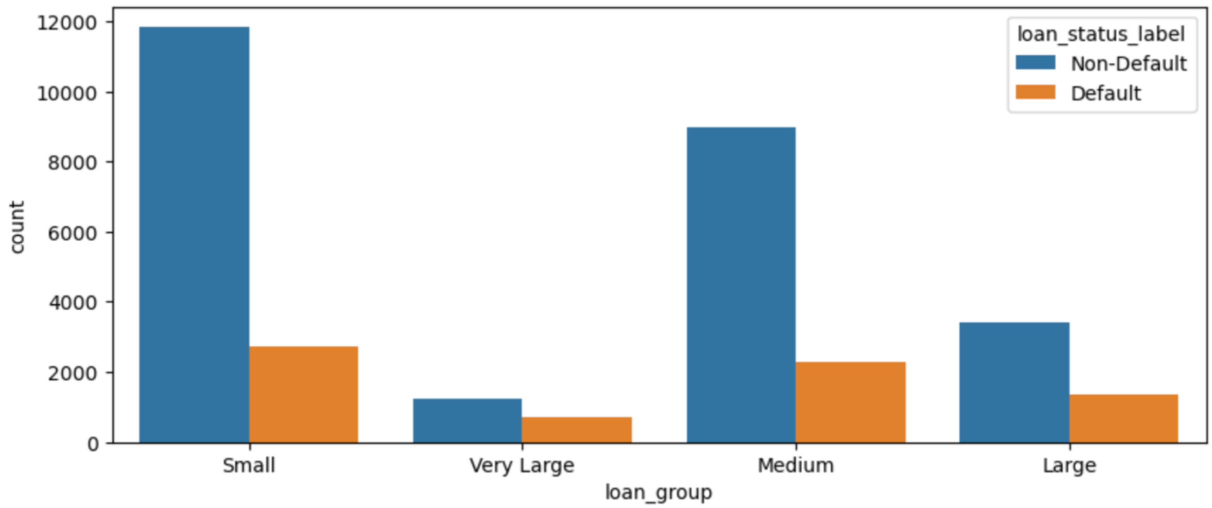
12. Which income group defaults the most?




```
income_group
High          8.483896
Low          33.237667
Medium       16.404431
Very High    10.208955
```

Low income applicants have the highest default rate (33.2%), which is 4x higher than the high income applicants (8.5%). Income is a strong predictor of default risk - lower income borrowers strugle significantly more to repay loans.

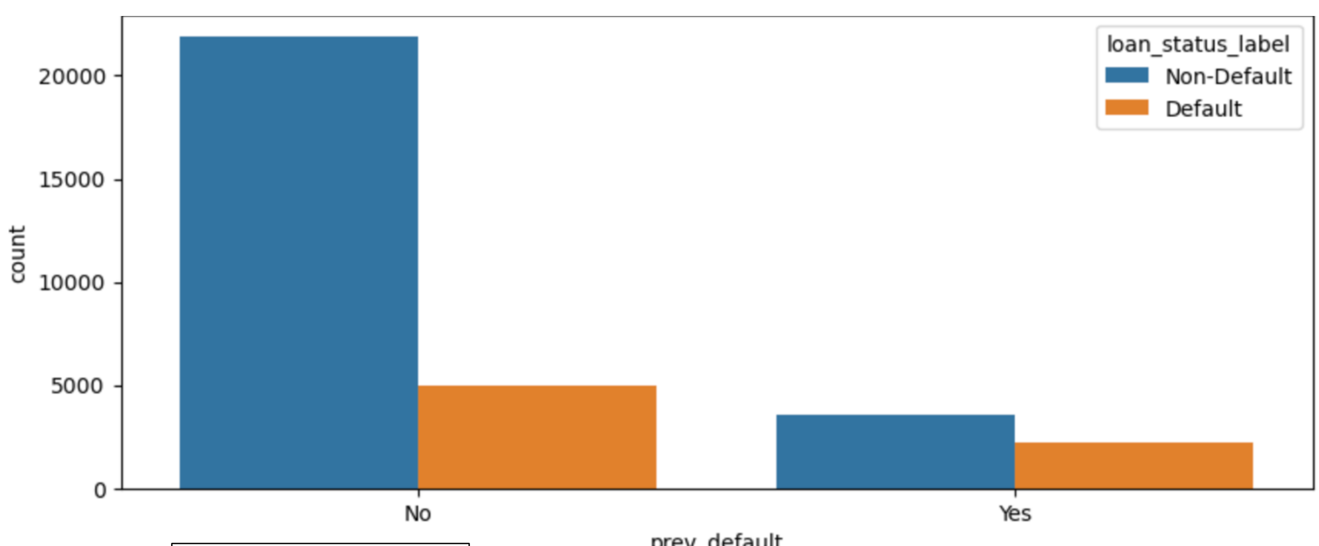
13. Which loan amount group defaults the most?




```
loan_group
Large          28.687840
Medium         20.361991
Small          18.805765
Very Large     36.076605
```

Very Large loans have the highest default rate (36%), followed by Large loan (28.7%). Small loans default the least (18.8%). This suggests borrowers taking large loans may be overextending their financial capaci ty.

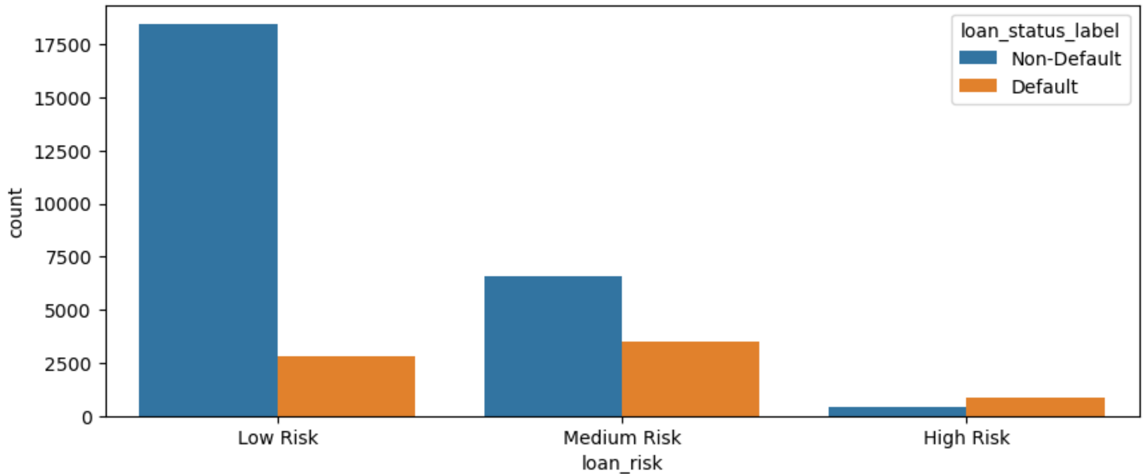
14. Does previous default history affect current default rate?




```
prev_default
No     18.401820
Yes    37.798293
```

Applicants with a previous default history are twice as likely to default again (37.8% vs 18.4%). Previous default is a strong indicator of future risk bank should exercise extra caution with these applicants.

15. Does loan risk group affect default rate?

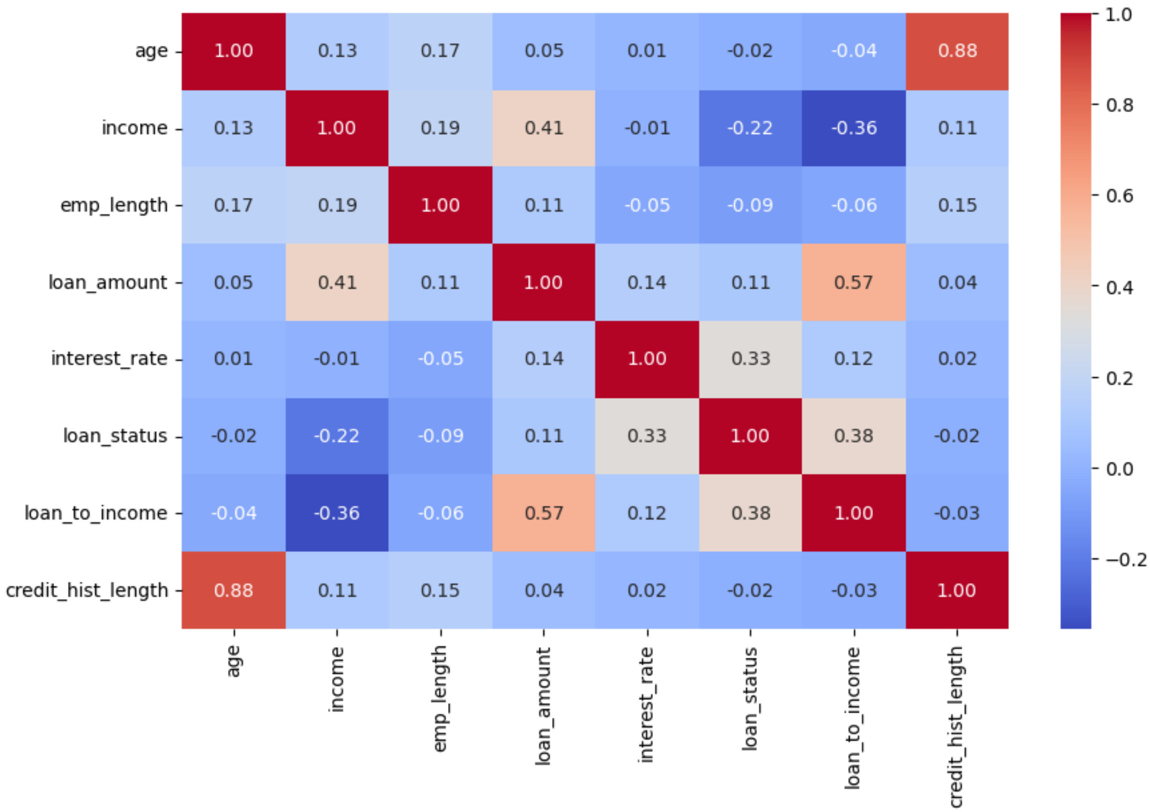



```
loan_risk
High Risk      67.297084
Low Risk       13.075654
Medium Risk    34.511166
```

High Risk loans (Grade E, F, G) have a staggering 67.3% default rate, compared to only 13.1% for Low Risk loans. Medium Risk sits at 34.5%. Loan risk group is clearly one of the strongest predictor of default behavior.

3.3. Correlation

16. How are numeric features correlated with loan status?



- Interest rate has the strongest positive correlation with loan status (0.33) - higher interest = more default.
- Loan to income ratio also strongly correlated (0.38) - borrowing more relative to income on increases default risk.
- Income has negative correlation with loan status (-0.22) - higher income = less default.
- Age and credit_hist_length are highly correlated (0.88) - older people naturally have longer credit history.
- Loan amount and loan to income are correlated (0.57) - larger loans take up more of income.

4. Key Insights & Recommendations

4.1 Key Insights

1. Overall default rate is 21.8% — 1 in 5 loans defaults.

2. Loan grade is the strongest predictor of default — Grade G has 98.4% default rate vs Grade A at 9.9%.

3. Low income borrowers default at 33.2% — nearly 4x higher than high income borrowers (8.5%).

4. Applicants with previous default history are twice as likely to default again (37.8% vs 18.4%).

5. Renters have the highest default rate (31.6%) — home owners are the most reliable borrowers (7.5%).

6. Debt Consolidation loans default the most (28.6%), while Venture loans default the least (14.8%).

7. Interest rate and loan-to-income ratio are the strongest numeric predictors of default.

4.2 Recommendations

1. Prioritize loan grade in approval decisions — avoid approving Grade E, F, G loans without additional collateral or guarantees.

2. Set stricter income verification requirements — low income applicants need closer scrutiny before loan approval.

3. Flag applicants with previous default history — require additional documentation or higher collateral from these applicants.

4. Introduce special monitoring for Debt Consolidation and Medical loans — these have consistently higher default rates.

5. Offer financial counseling to renters before approving large loans — home ownership indicates stronger financial stability.

5. Conclusion

This exploratory data analysis of the Credit Risk dataset reveals several critical factors driving an overall default behavior. With an overall default rate of 21.8%, credit risk management is a key priority for financial institutions.

The analysis identifies loan grade, income level, previous default history, and home ownership as the strongest predictors of default risk. High risk grade loans (E, F, G) show extremely high default rates, confirming that credit grade is the most reliable indicator of repayment behavior.

On the other hand, high income borrowers and home owners demonstrate significantly lower default rates, suggesting that financial stability plays a crucial role in loan repayment capability.

The insights derived from this analysis can guide credit risk teams to make more informed, data-driven loan approval decisions — ultimately reducing default rates and improving portfolio quality.

A Tableau dashboard has been developed alongside this report to enable interactive exploration of default patterns across different borrower segments and loan characteristics.